

# Abstract

# Acknowledgements

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# Chapter 1

## Strategic Vision and Project Scope

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## 1.1 General Introduction

Digital transformation has changed how educational institutions manage teaching, communication, assessment, and academic follow-up. Despite this shift, many institutions still rely on fragmented combinations of messaging applications, file-sharing tools, video-conferencing platforms, and manually coordinated assessment processes. Such fragmentation reduces continuity in the learning experience, increases administrative overhead, and weakens the institutional ability to manage educational workflows within one controlled environment.

Eduverse is an integrated educational platform developed to address this problem through a unified, multi-organization, role-based environment. Its main implemented client is a web application, supported by an MVP mobile companion application that extends access to selected academic workflows for students and teachers. The platform is centered on the class as the primary digital workspace in which materials, assignments, examinations, results, notifications, live sessions, academic communication, and intelligent support services can be managed in a coordinated manner.

At a high level, Eduverse combines a modern web platform with supporting cloud and real-time services in order to provide a realistic and extensible educational system. The web client is built using Next.js, React, TypeScript, Tailwind CSS, and Supabase, while the mobile companion is built using Expo, React Native, and TypeScript. These technologies are mentioned here only as an introductory overview; detailed technical analysis and implementation discussion are reserved for later chapters.

## 1.2 Problem Statement

Educational institutions often manage closely related academic tasks through separate systems that are not designed to operate as one coherent platform. Materials may be distributed through one service, communication may take place in another, live teaching may depend on a third platform, and assessment records may be maintained elsewhere. As a result, students are forced to move between disconnected interfaces, while teachers and administrators struggle to maintain consistent oversight of participation, submissions, communication, and evaluation.

This problem becomes more serious in institutions that must manage mul-

tiple organizations, classes, and user roles under different access conditions. A practical educational platform must support not only learning activities, but also institutional control, membership management, and role-based access across different organizational contexts. Without these capabilities, digital education systems tend either to become rigid and limited or to become too loosely structured to support reliable academic administration.

A further challenge concerns platform coverage across devices. Rich authoring, administration, and class management workflows are best handled through a full web environment, whereas frequent day-to-day actions such as checking classes, following notifications, viewing assignments, opening materials, joining live sessions, and reading messages must also be accessible on mobile devices. This creates a need for an educational platform centered on a comprehensive web environment, supported by a practical mobile companion rather than a simplistic one-client-for-all approach.

### 1.3 Project Motivations

The development of Eduverse is motivated by the need to provide a unified academic environment in which essential educational workflows can be carried out without fragmentation. A coherent class-centered system can improve the student experience, reduce operational burden for teachers and administrators, and strengthen institutional visibility over communication, assessment, and academic progress.

The project is also motivated by the growing demand for flexible educational infrastructure. Modern institutions often require platforms that can support multiple organizations, distinct user roles, synchronous and asynchronous learning activities, and selective feature enablement within one extensible environment. Eduverse responds to this need by combining institutional control with practical support for daily teaching and learning workflows.

From an academic perspective, Eduverse is also motivated by its value as a software engineering project. It provides a realistic setting in which requirements analysis, interface planning, system integration, data management, real-time communication, AI-assisted services, and technical documentation can be studied and implemented within one coherent graduation project context.

## 1.4 Strategic Objectives

At the strategic level, Eduverse aims to establish an integrated educational platform whose primary implemented client is a web application, complemented by an MVP mobile companion application, in order to centralize essential academic workflows for organizations, teachers, and students within one role-aware and extensible system.

The main strategic objectives of the project are as follows:

- To centralize core educational workflows such as class management, materials, assignments, examinations, results, notifications, communication, and live sessions within one coordinated platform.
- To support multi-organization operation and role-based access for organization owners, organization admins, teachers, and students.
- To provide a primary web-based environment capable of supporting both institutional administration and day-to-day academic activity.
- To extend selected high-frequency academic workflows to a mobile MVP companion application without claiming full feature parity with the web platform.
- To incorporate distinctive educational capabilities such as an AI Agent and an integrated educational IDE as part of the platform's broader academic value.
- To organize data and services in a maintainable and scalable manner suitable for educational institutions with different operational needs.
- To produce a graduation project that is both academically meaningful and practically relevant as a modern educational software platform.

## 1.5 Educational and Professional Goals for the Team

Eduverse also serves clear educational and professional goals for the project team. From an educational standpoint, it provides an opportunity to apply software engineering concepts to a realistic problem domain rather than to

an isolated classroom exercise. The team must analyze institutional needs, define system boundaries, organize requirements, structure user roles, and translate these decisions into a functional digital platform.

From a professional standpoint, the project strengthens practical experience in requirements analysis, frontend development, backend integration, database usage, real-time service integration, collaborative development workflows, and technical documentation. It therefore contributes not only to the production of software, but also to the development of disciplined engineering practice, teamwork, and academic reporting skills.

## 1.6 Project Scope and Boundaries

The scope of Eduverse in this dissertation is defined around the implemented web platform and the existing mobile MVP companion application. Chapter 1 is limited to introducing the problem context, strategic direction, main user groups, system boundaries, and general dissertation structure. Detailed requirements, architecture, database design, implementation decisions, and testing evidence are reserved for later chapters.

### 1.6.1 System Boundaries and User Roles

Eduverse operates within a multi-organization educational context and therefore distinguishes clearly between administrative, instructional, and learner responsibilities. At the institutional level, the platform supports organization owners and organization admins who manage organizations, user access, and operational control. At the instructional level, teachers manage class activities such as materials, assignments, live sessions, examinations, and academic follow-up. At the learner level, students access the classes to which they belong in order to study materials, follow announcements, submit work, join sessions, communicate, and review academic outcomes.

These role distinctions define important system boundaries. Eduverse is not a generic social platform in which all users have the same functional position. Instead, access is governed by organizational membership, class context, and academic responsibility. This role-aware structure supports accountability, controlled visibility, and more reliable institutional management.

## 1.6.2 Functional Scope

Within its current implemented scope, Eduverse includes authentication, multi-organization access, class management, materials, assignments, exams, results, notifications, live sessions, an AI Agent, and an integrated educational IDE. Together, these functions define the platform as more than a basic learning content portal. Eduverse is intended to support both academic management and active instructional workflows within one system.

At the same time, Chapter 1 remains introductory in nature. It does not attempt to document detailed database structures, API behavior, or low-level internal logic. Those aspects are part of the later analytical and technical chapters and lie outside the present scope.

## 1.6.3 Web and Mobile Client Scope

The main implemented Eduverse client is the web application. It provides the widest operational coverage of the platform, including organization-aware navigation, role-based access, class management, materials, assignments, exams, results, notifications, live sessions, AI Agent workflows, and the integrated educational IDE. For this reason, the web application should be regarded as the core working environment of the system.

The mobile application has a more focused scope. It is an MVP companion application intended to extend access to selected academic workflows such as authentication, dashboard viewing, classes, assignments, notifications, chat, materials, and live sessions. Although it is connected to the same broader platform environment, it is not yet feature-complete relative to the web application and should therefore be described as a companion application rather than a replacement for the primary platform.

## 1.6.4 Competitive Features and Technological Innovation

Eduverse is distinguished by the combination of features it brings together in one educational platform. A major distinguishing point is its multi-organization, role-based structure, which allows the same system to support organizational control, instructional management, and student participation in a coordinated way. The platform also integrates live sessions, an

AI Agent, and an educational IDE within the same broader academic environment rather than treating them as isolated external utilities.

Another important feature is the platform’s attention to examinations and assessment integrity. Eduverse does not treat evaluation as a peripheral task, but as an integral part of the educational workflow that must be managed within a controlled digital environment. In addition, the system deliberately combines a comprehensive web platform with an MVP mobile companion application, thereby balancing rich institutional workflows with accessible everyday mobile use.

From a technological perspective, Eduverse also reflects a deliberate service separation strategy. Structured and frequently queried academic data are stored through Supabase PostgreSQL, large educational files are handled through AWS S3, and real-time live-session interaction is supported through LiveKit. This separated storage and service model strengthens the platform’s practicality for an educational environment that must manage both relational records and larger real-time or file-based workloads.

## 1.7 Report Organization

This dissertation is organized as follows. Chapter 2 presents the literature review and theoretical framework. Chapter 3 describes the adopted methodology. Chapter 4 covers system analysis and requirements. Chapter 5 presents system design. Chapter 6 discusses software and engineering implementation. Chapter 7 presents testing and evaluation. Chapter 8 addresses future work and planned extensions, followed by the conclusion and appendices.

# Chapter 2

## Literature Review and Theoretical Framework

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| Theoretical and Technical Foundations of Eduverse .....              | 2.6, p. 12 |

## 2.1 Introduction

This chapter presents the theoretical background that frames Eduverse as an integrated educational platform. It reviews the shift from traditional classroom management toward digital learning systems, highlights the limitations of fragmented educational tools, and situates Eduverse within the broader need for unified academic ecosystems. The chapter also discusses selected concepts from artificial intelligence in education, synchronous virtual learning, and secure role-aware platform design in order to establish the conceptual basis for the later analysis, design, and implementation chapters.

## 2.2 Evolution of Digital Education Platforms

### 2.2.1 From Traditional Classroom Management to Digital Learning Systems

Educational management has historically depended on face-to-face coordination, manual attendance records, printed materials, and locally maintained assessment processes. As educational institutions expanded in scale and digital communication became more common, Learning Management Systems emerged to centralize course materials, announcements, assignment handling, and academic tracking within a unified digital environment [1, 2].

The move toward digital learning systems did not simply replace paper-based administration with electronic storage. It introduced new expectations about continuous access, asynchronous participation, online communication, and traceable academic workflows. Modern educational platforms are therefore judged not only by their capacity to store materials, but also by their ability to support coordination, interaction, assessment, and institutional oversight across different users and contexts [11].

### 2.2.2 Limitations of Fragmented Learning Management Tools

Although digital tools have become common in education, many institutions still depend on separate services for class management, communication, file sharing, video sessions, assessment, and student support. This fragmented model creates operational discontinuity because activities that belong to the



same academic process are distributed across unrelated systems. As a result, students must move repeatedly between platforms, while teachers and administrators face a reduced ability to maintain clear academic oversight [11, 2].

Fragmentation also affects consistency and accountability. When communication, submissions, live sessions, and evaluation records are distributed across multiple tools, it becomes more difficult to maintain a coherent academic history, enforce clear access boundaries, and manage workflows in a structured way. From a system-design perspective, the limitation is not merely technical; it is also organizational, because fragmented tools weaken the integrity of the educational environment itself.

## 2.3 Research Gap and Proposed Integrated Learning Ecosystem

A clear gap exists between the broad range of activities required in modern education and the way those activities are often supported by disconnected systems. Institutions may rely on one tool for class coordination, another for messaging, another for live teaching, another for assignments, and still others for assessment, AI support, or programming practice. While each tool may be effective in isolation, the overall experience remains fragmented and operationally inefficient.

Eduverse addresses this gap by proposing an integrated learning ecosystem rather than a collection of separate utilities. Its core idea is that class management, communication, materials, assignments, live sessions, examinations, AI-assisted support, and programming practice should be coordinated within one platform governed by organizational structure and role-based access. In this sense, Eduverse is positioned as a response to fragmentation through integration, controlled scope, and educational workflow continuity.

This research direction is especially relevant because the platform does not attempt to treat every educational function as identical in weight or usage. Instead, it distinguishes between the main implemented web environment and the MVP mobile companion application, and between core academic workflows and specialized extensions such as AI support and the educational IDE. That balance between integration and scope control is central to the academic identity of the project.

## 2.4 Artificial Intelligence in Education

Artificial intelligence in education is commonly discussed as a means of augmenting teaching and learning rather than replacing teachers or institutional judgment [8, 6]. In educational settings, AI systems are valuable when they improve access to explanations, support learning continuity, reduce routine workload, and provide structured assistance without undermining academic responsibility.

### 2.4.1 Context-Aware Learning Assistance

One of the most important ideas in AI-supported education is that assistance is more useful when it is tied to the learner’s actual academic context. Generic responses are often less effective than support that takes into account course materials, class discussion, ongoing tasks, and instructional expectations. Context-aware assistance can therefore improve relevance, reduce confusion, and support more focused academic interaction [6].

In relation to Eduverse, this concept is particularly important because the AI Agent is not conceived as an isolated chatbot detached from the broader learning environment. Its academic value depends on being situated within the class context and used as a support layer for understanding, clarification, and workflow assistance rather than as a substitute for teaching.

### 2.4.2 Material Summarization and Learning Support

Material summarization represents a practical form of AI assistance in education. Long documents, learning materials, and instructional resources can be difficult for students to process efficiently, especially when they are working under time constraints or trying to review several academic topics in parallel. AI-assisted summarization can support comprehension by highlighting main points, reducing information overload, and helping learners return more quickly to the central ideas of a topic [8].

This kind of support is most academically appropriate when it complements reading rather than replacing it. For that reason, summarization should be understood as a learning aid that improves navigation and revision rather than as an alternative to engagement with the full material.

### 2.4.3 Assessment and Feedback Support

AI can also support assessment-related workflows by assisting with draft guidance, formative feedback, and structured academic support around tasks and examinations. In theory, such support can help learners prepare, reflect on their work, and identify areas that require improvement [6]. However, educational literature also emphasizes that AI use in assessment must be governed carefully so that it supports learning without weakening academic integrity or teacher responsibility [8, 6].

For Eduverse, this means AI should be treated as a support mechanism within the assessment environment, not as an autonomous evaluator that replaces human judgment. This distinction is especially important in educational systems that must preserve trust, fairness, and instructor oversight.

## 2.5 Benchmarking: Fragmented Tools vs. Integrated Eduverse Approach

Table 2.1 summarizes the difference between a fragmented educational-tool model and the integrated approach proposed by Eduverse.

## 2.6 Theoretical and Technical Foundations of Eduverse

### 2.6.1 Web-Based Educational Platforms

Web-based educational platforms remain the strongest medium for full academic management because they support richer interfaces, broader administrative workflows, and deeper integration across organizational functions. They are especially suitable for tasks such as class configuration, materials management, assessment setup, and detailed instructional coordination.

For Eduverse, the web platform represents the primary implemented client because it can accommodate both institutional management and complex daily teaching workflows in a way that smaller device environments cannot easily match. This makes the web environment foundational not only technically, but also pedagogically and operationally.

Table 2.1: Fragmented Educational Tools vs. Integrated Eduverse Approach

| <b>Educational Need</b>   | <b>Fragmented Tool Approach</b>                                                                                                        | <b>Integrated Eduverse Approach</b>                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Class management          | Class records, memberships, and academic coordination are handled through separate administrative tools or partially manual processes. | Class organization is managed within one platform through organization-aware and role-based structures.    |
| Communication             | Messaging and announcements are distributed across external chat or notification tools.                                                | Communication remains connected to the academic environment and class context.                             |
| Live sessions             | Synchronous teaching depends on external meeting services disconnected from course workflows.                                          | Live sessions are treated as part of the broader educational platform and linked to class participation.   |
| Materials and assignments | Learning resources and tasks are often stored, shared, and submitted through different services.                                       | Materials and assignments are coordinated within the same academic workspace.                              |
| AI support                | AI tools are usually external and not directly tied to the learner's course context.                                                   | AI support is positioned within the class environment as contextual academic assistance.                   |
| Programming practice      | Coding activities, when needed, often require separate platforms or disconnected lab tools.                                            | Programming practice can be supported through an integrated educational IDE within the platform ecosystem. |
| Assessment                | Examinations and grading may rely on detached tools with limited continuity or oversight.                                              | Assessment workflows are integrated into the same role-aware educational system.                           |

## 2.6.2 Mobile Companion Learning Applications

Mobile educational applications play a different but still important role in digital learning environments. Their strength lies in quick access, timely participation, and support for routine high-frequency actions such as checking notifications, reviewing assignments, opening materials, and joining live sessions. As a result, they are especially useful as companion systems for daily academic engagement.

In Eduverse, the mobile client should therefore be understood as an MVP mobile companion rather than as a complete replacement for the main platform. Its purpose is to extend accessibility to core user-facing workflows while the more comprehensive management and authoring activities remain centered in the web application.

## 2.6.3 Real-Time Virtual Classrooms

Synchronous learning environments are an important part of modern educational theory because they strengthen immediacy, interaction, and instructional presence. When live classrooms are integrated into the broader academic platform rather than isolated in external services, continuity between teaching, communication, and follow-up becomes easier to maintain [5].

This foundation is relevant to Eduverse because live sessions are not treated as a separate product layer. Instead, they are part of the integrated learning environment and serve the broader goal of linking synchronous participation with the rest of the class workflow.

## 2.6.4 Integrated Coding Environments for Programming Education

Programming education often requires an environment in which learners can write code, inspect results, practice problem solving, and remain connected to the instructional context of the class. When coding practice is separated entirely from the educational platform, students may lose continuity between learning materials, teacher guidance, assignments, and practical execution.

An integrated coding environment addresses this problem by placing programming practice within the same educational ecosystem as the rest of the academic workflow. For Eduverse, the integrated educational IDE is therefore not merely a technical add-on, but part of the platform's broader pedagogical

direction, especially for programming-related classes and technical learning settings.

### **2.6.5 Security, Roles, and Data Management**

Trustworthy educational platforms depend on controlled access, clear role distinctions, and dependable data management. Role-Based Access Control is especially relevant in this context because it supports structured separation between administrative, teaching, and learner permissions [10, 4]. In educational systems, such distinctions are necessary not only for security, but also for accountability and appropriate institutional governance.

Data management is equally foundational. Educational systems must maintain reliable records, protect assessment-related information, and support different types of content ranging from structured academic data to larger learning materials and live-session interaction. From a theoretical perspective, this requires a design orientation that treats security, data organization, and role awareness as core educational infrastructure rather than as secondary technical concerns. This view is consistent with Eduverse, where access control, data separation, and institutional structure are part of the platform's academic trust model.

# Chapter 3

## Methodology

*TODO: This chapter outline is reserved for a later documentation step.*

## **3.1 Introduction**

## **3.2 Adopted Development Methodology**

### **3.2.1 Iterative / Agile Development**

### **3.2.2 Justification of Technology Choices**

## **3.3 Development Phases**

### **3.3.1 System Planning and Requirements**

### **3.3.2 UI/UX and Frontend Development**

### **3.3.3 Backend and Database Development**

### **3.3.4 Exams, Live Sessions, and Communication Features**

### **3.3.5 Testing and Refinement**

## **3.4 Modeling and Documentation Approach**

## **3.5 Summary**



## Chapter 4

# System Analysis and Requirements

*TODO: This chapter outline is reserved for a later documentation step.*

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### 4.8.4 Assessment Workflows

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# Chapter 5

## System Design

*TODO: This chapter outline is reserved for a later documentation step.*

- 5.1 Introduction
- 5.2 Architectural Design Overview
- 5.3 Data and Storage Design
- 5.4 Interface and Interaction Design
- 5.5 Integration and Security Design

# Chapter 6

## Software and Engineering Implementation

*TODO: This chapter outline is reserved for a later documentation step.*

### 6.1 Introduction

### 6.2 Web Application Implementation

### 6.3 Mobile Companion Implementation

### 6.4 Backend and Service Integration

### 6.5 Engineering Challenges and Resolutions

# Chapter 7

## System Testing and Evaluation

*TODO: This chapter outline is reserved for a later documentation step.*

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### 7.2 Testing Strategy

### 7.3 Functional Testing

### 7.4 Non-Functional Evaluation

### 7.5 Discussion of Results

# Chapter 8

## Future Work and Planned Features

*TODO: This chapter outline is reserved for a later documentation step.*

### 8.1 Introduction

### 8.2 Mobile Expansion

### 8.3 AI Agent and IDE Enhancement

### 8.4 Scalability and Institutional Growth

## Chapter 9

## Conclusion

*TODO: Write the final conclusion after the main report chapters are completed.*

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**Appendix A**

**Appendices**